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AI/ML Engineer · NLP Researcher · Data Scientist

PROFESSIONAL SUMMARY

AI/ML engineer and researcher. Three years writing production software, plus four spent building and shipping AI systems and data science based software that real people use. First-author on three ACL-venue NLP papers (SemEval-2026, BEA 2026). I build LLM systems end to end. Agentic workflows, RAG pipelines, models squeezed onto edge hardware. Most of it came out of cross-university work, deployed for users in healthcare, education, and climate policy.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Go, Dart, Java, C/C++, Rust, SQL (PostgreSQL)

ML & NLP: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, fine-tuning, contrastive learning, knowledge distillation, model optimization, embeddings, multilingual NLP

LLM & Agentic AI: LangGraph, LangChain, LangSmith, Model Context Protocol (MCP), RAG, prompt engineering, OpenAI API, ChromaDB, agentic workflows

Web & Backend: React, Next.js, Flask, Node.js, Vite, REST API, gRPC, microservices, API gateways

Infrastructure: Docker, Kubernetes, Git, Linux, nginx, Google Cloud Platform, CI/CD, distributed tracing, observability

Spoken: Indonesian (native), English (professional, TOEFL ITP 587), German (B1), Javanese (conversational)

EXPERIENCE

Researcher and Lead Software Engineer

Feb. 2022 – Present

Mantera Studio Research Lab (PT. BSM)

Jakarta, Indonesia

- Published first-author at SemEval-2026 Task 11, finishing top-5 across all subtasks on multilingual formal reasoning via deterministic parsing and structural abstraction of syllogisms into canonical logical forms.
- Published first-author at BEA 2026 Shared Task, ranking top-15 across all L1 groups (14th Spanish, 11th German, 7th Mandarin) with a Ridge ensemble over XLM-RoBERTa and mDeBERTa using ordinal soft contrastive learning.
- Lead engineering across the lab's production AI portfolio, directing system architecture and shipping customer-facing LLM applications in collaboration with Queen's University Belfast, UGM, and Deakin University.

Research Assistant, Machine Learning Systems

Feb. 2026 – Present

Universitas Gadjah Mada

Yogyakarta, Indonesia

- Develop early-exit training and inference methods for energy-efficient, low-latency model deployment on resource-constrained edge hardware (NVIDIA Jetson Orin) across 7W–25W power budgets.

Research Assistant, Software Engineering Lab

Mar. 2025 – Oct. 2025

Universitas Gadjah Mada

Yogyakarta, Indonesia

- Led ML workstream for a national grant: delivered PHQ-aligned LLM screening with deep-learning facial affect analysis, evaluated on 302 samples, and presented results to university stakeholders and grant reviewers.
- Built LLM-based NLP tooling for requirements engineering, adding pragmatic and semantic quality checks for user-story specifications that extended prior automated QUS analysis.

Web Developer

2019 – 2022

Mastej Studio

Malang, Indonesia

- Built and shipped What's On Malang, a React/Vite tourism platform (400+ DAU, 100 leads/day), and the Begawan apartment marketing site with 360-degree virtual tours (10 daily inquiries); delivered demos and prototypes directly to regional clients.
URLS: whatsonmalang.com · begawan-apartment.com
- Co-developed Janur, a digital wedding-invitation site with built-in attendance tracking. Helping build 7 custom invitations for weddings
URL: janur.wedding
- Helped run Pasar Santai, a local market event held annually in malang every ramadhan, as logistics officer and web developer.
URL: pasarsantai.themastej.com

PUBLICATIONS

ITLC at SemEval-2026 Task 11: Normalization and Deterministic Parsing for Formal Reasoning in LLMs. First author (Mantera Studio), SemEval-2026 (ACL). Top-5 across all subtasks, outperforming fine-tuning and activation-level baselines.

URL : arxiv.org/abs/2603.02676

Improving Lexical Difficulty Prediction with Context-Aligned Contrastive Learning and Ridge Ensembling. First author (Mantera Studio), BEA 2026 Shared Task (ACL). Top-15 across all L1 groups (14th ES, 11th DE, 7th ZH).

URL : arxiv.org/abs/2605.08950

Aligning Implied Statements for Implicit Hate Speech Generalizability. First author, under review (ACL ARR). +4–8% out-of-domain F1 across 3 benchmarks via triplet contrastive learning with context-bounded semi-hard negatives.

URL : openreview.net/pdf?id=pgwGGyy77k

On-Edge Device Optimization Using Multiple Classification Method for a Cat and Dog Audio Classifier. First author, IEEE. 95.2% accuracy, 3 ms latency, 3.8 KB memory on Arduino Nano 33 BLE.

URL : ieeexplore.ieee.org/abstract/document/10933173

Enhancing Automated User Story Quality Assessment with Large Language Models. Co-author, ICODSE 2025.

URL : <https://ieeexplore.ieee.org/abstract/document/11351886>

SELECTED PROJECTS

Dyslexic Reader (Mantera Studio × UGM × Queen’s University Belfast) · *Dart, Next.js, Kubernetes, Python, gRPC, webhook*

URL : dyslexic.app

2026

- Built LLM-powered dyslexia support app for a Mantera Studio collaboration with Queen’s University Belfast and UGM, deployed on Kubernetes with Dart mobile and Next.js web frontends across multiple VPS.
- Implemented a real-time font-rendering dyslexia lens, LLM summarization and text-professionalization, and an AHRQ-aligned screening protocol for clinical alignment.

Mental Health Screening Platform (Mantera Studio) · *Flask, LangGraph, LangChain, PyTorch,webhook,LiteRT*

URL : mentalhealth.laptopmerahputih.id

Nov. 2025

- Shipped LLM-driven PHQ + CPAS screening workflows with LangGraph agentic orchestration, serving 40+ concurrent sessions with stable uptime.
- Integrated deep-learning facial affect capture; cut p95 latency and serving cost 90% via local response batching and request queuing, monitored with LangSmith.

RAG Chatbots: TASYA & IndoClimate (Mantera Studio × UGM, Deakin) · *Flask, React, ChromaDB, LangGraph*

URLs : tasya.dtmiumg.id · chat.indoclimate.id

2025

- Built WhatsApp webhook and REST RAG services over institutional docs and Indonesian climate legislation; indexed ~2,000 pages across 7 documents into ChromaDB.
- Orchestrated multi-step retrieval and generation via LangGraph; added an LLM summarization pipeline for long legal documents to improve vector matching, traced with LangSmith.

SOC Dashboard (Mantera Studio · under NDA) · *Vite, Redux Toolkit, Go, LangGraph, LangSmith*

2024

- Refactored frontend to domain-driven design, cutting redundant code 70%; integrated server telemetry ingestion, distributed tracing, audit logging, and LLM-based log summarization.

Sijinak Attendance System (Mantera Studio) · *gRPC, FFI, Dart, Python, C*

URL : github.com/airlangawicaksono/sijinak_win, <http://sim.man2yogyakarta.sch.id/>

2025

- Built RFID-based attendance system integrating Hikvision hardware via Foreign Function Interface to a native C SDK for hardware interaction of the RFID device used; exposed a microservice layer over gRPC with a Dart frontend.

EDUCATION

Universitas Gadjah Mada
Master of Artificial Intelligence

Yogyakarta, Indonesia
Feb. 2024 – Oct. 2025

Institut Teknologi Sepuluh Nopember
B.A.Sc. Computational and Applied Mathematics

Surabaya, Indonesia
Aug. 2018 – Nov. 2022